

Jongin Lim, Ph.D.

PROFILE	Staff Engineer AI Center, Samsung Electronics Tel: +82-10-7429-7508 E-mail: jonny.lim@samsung.com	
WORK EXPERIENCE	AI Center, Samsung Electronics Staff Engineer at Process/Equipment AI Lab	Jan. 2025 – Now
	Samsung Advanced Institute of Technology Staff Engineer at Machine Learning TU	Mar. 2022 – Dec. 2024
EDUCATION	Seoul National University Integrated M.S./Ph.D. in Electrical and Computer Engineering (Advisor: Prof. Jin Young Choi)	Sep. 2016 – Feb. 2022
	Seoul National University B.S. in Electrical and Computer Engineering	Mar. 2011 – Feb. 2016
PUBLICATIONS	(C: Conference paper, J: Journal paper)	
	[C12] Jongin Lim , Sucheol Lee, Daeho Um, Sung-Un Park, and Jinwoo Shin, “PRIME: Deep Imbalanced Regression with Proxies”, The 42nd International Conference on Machine Learning (ICML), 2025.	
	[C11] Daeho Um, Sunoh Kim, Jiwoong Park, Jongin Lim , Seong Jin Ahn, Seulki Park, “Propagate and Inject: Revisiting Propagation-Based Feature Imputation for Graphs with Partially Observed Features”, The 42nd International Conference on Machine Learning (ICML), 2025.	
	[C10] Daeho Um, Jongin Lim , Sunoh Kim, Yuneil Yeo, and Yoonho Jung, “Spreading Out-of-Distribution Detection on Graphs”, The 13th International Conference on Learning Representations (ICLR), 2025.	
	[C9] Chanho Ahn, Kikyung Kim, Ji-won Baek, Jongin Lim , and Seungju Han, “Sample-wise Label Confidence Incorporation for Learning with Noisy Labels”, IEEE International Conference on Computer Vision (ICCV), 2023.	
	[C8] Jongin Lim , Youngdong Kim, Byungjai Kim, Chanho Ahn, Jinwoo Shin, Eunho Yang, and Seungju Han, “BiasAdv: Bias-Adversarial Augmentation for Model Debiasing”, IEEE Computer Vision and Pattern Recognition (CVPR), 2023.	
	[C7] Jongin Lim , Sangdoo Yun, Seulki Park, and Jin young Choi, “Hypergraph-Induced Semantic Tuple Loss for Deep Metric Learning”, IEEE Computer Vision and Pattern Recognition (CVPR), 2022.	
	[C6] Seulki Park, Jongin Lim , Younghan Jeon, and Jin young Choi, “Influence-balanced Loss for Imbalanced Visual Classification”, IEEE International Conference on Computer Vision (ICCV), 2021.	
	[C5] Jongin Lim , Daeho Um, Hyung Jin Chang, Dae Ung Jo, and Jin Young Choi, “Class-Attentive Diffusion Network for Semi-Supervised Classification”, The 35th AAAI Conference on Artificial Intelligence (AAAI), 2021.	
	[C4] Youngmin Ro, Jongwon Choi, Dae Ung Jo, Byeongho Heo, Jongin Lim , and Jin Young Choi, “Backbone Cannot Be Trained at Once: Rolling Back to Pre-trained	

Network for Person Re-identification”, The 33rd AAAI Conference on Artificial Intelligence (**AAAI**), 2019.

- [C3] **Jongin Lim**, Hyung Jin Chang, and Jin Young Choi, “PMnet: Learning of Disentangled Pose and Movement for Unsupervised Motion Retargeting”, British Machine Vision Conference (**BMVC**), 2019.
- [J2] **Jongin Lim**, Youngjoon Yoo, Byeongho Heo, and Jin Young Choi, “Pose Transforming Network: Learning to Disentangle Human Posture in Variational Auto-encoded Latent Space”, **Pattern Recognition Letters**, 2018.
- [C2] Jiyeoup Jeong, Hawook Jeong, **Jongin Lim**, Jongwon Choi, Sangdoo Yun, and Jin Young Choi, “Selective Ensemble Network for Accurate Crowd Density Estimation”, The 24th International Conference on Pattern Recognition (**ICPR**), 2018.
- [J1] Kimin Yun, **Jongin Lim**, and Jin Young Choi, “Scene Conditional Background Update for Moving Object Detection in a Moving Camera”, **Pattern Recognition Letters**, 2017.
- [C1] Kimin Yun, **Jongin Lim**, Sangdoo Yun, Soo Wan Kim, and Jin Young Choi, “Attention-Inspired Moving Object Detection in Monocular Dashcam Videos”, The 23rd International Conference on Pattern Recognition (**ICPR**), 2016.

PATENTS

Method and Device with Image-Difference Reduction Preprocessing [[Google patent](#)]

- KR20250020150A, South Korea (Publication: 2025-02-11)
- US20250045884A1, United States (Publication: 2025-02-06)
- TW202507651A, Taiwan (Publication: 2025-02-16)
- EP4502936A3, European Patent Office (Publication: 2025-05-14)
- CN119444583A, China (Publication: 2025-02-14)

Method and Electronic Device with Adversarial Data Augmentation [[Google patent](#)]

- US20240152764A1, United States (Publication: 2024-05-09)
- JP2024066469A, Japan (Publication: 2024-05-15)
- EP4365777A1, European Patent Office (Publication: 2024-05-08)

Method and Device with Defect Detection [[Google patent](#)]

- KR20240064412A, South Korea (Publication: 2024-05-13)
- US20240153070A1, United States (Publication: 2024-05-09)
- CN117994200A, China (Publication: 2024-05-07)
- EP4365834A1, European Patent Office (Publication: 2024-05-08)
- JP2024068105A, Japan (Publication: 2024-05-17)

(S: Samsung project, G: Graduate research project)

- [S3] **Multi-modal Virtual Metrology** Mar. 2024 – Now
I am contributing to the development of an ML-based virtual metrology system to predict critical dimension (CD) values in semiconductor manufacturing. Since direct measurement of CD for all wafers is costly, this project leverages multi-modal data (*e.g.*, FDC, OES, and process parameters) to enable accurate CD estimation, improving efficiency and reducing metrology costs.
- [S2] **Multi-modal Foundation Model for Semiconductor** Mar. 2024 – Now
I am contributing to the development of a foundation model leveraging multi-modal semiconductor data to support various applications across the semiconductor manufacturing process at Samsung. This project aims to build a scalable and transferable model by integrating diverse data sources, including process parameters, metrology data, and equipment sensor readings, enabling robust and generalizable AI solutions for semiconductor manufacturing.
- [S1] **Bright Field (BF) Inspection** Mar. 2022 – Feb. 2024
Contributed to the development of a BF inspection system for high-speed defect detection in semiconductor manufacturing. This project focused on designing a robust and fine-grained ML model capable of detecting subtle variations, ensuring precise defect identification and improved inspection accuracy.
- [G4] **Person Re-ID for Multi-camera System** May. 2020 – Nov. 2020
Sponsored by **LG Electronics**. The goal is to develop a person re-identification algorithm that matches a query image of pedestrian to images in pedestrian image DB. We have developed the efficient person re-identification algorithm and evaluated on Market-1501 and DukeMTMC.
- [G3] **Moving Object Detection for Future Soldier** May. 2018 – Aug. 2019
Sponsored by **LIG Nexone**. The goal of this project is to develop moving object detection algorithm to detect moving objects in real time in high-resolution images captured in an environment where the camera is not fixed.
- [G2] **Development of Multi-modal Sensor-based Outdoor Surveillance Systems** Apr. 2017 – Nov. 2018
Sponsored by **Ministry of S&ICT**. The goal is to develop multi-modal sensor-based object detection, information fusion, and context-aware recognition technologies for intelligent learning and recognition of outdoor unmanned security robots in the dynamic surveillance environment, and to develop a multi-agent-based intelligent information analysis platform.
- [G1] **Development of Predictive Visual Intelligence Technology** Sep. 2016 – May. 2019
Sponsored by the **ICT R&D program of MSIP/IITP**. The goal of this project is to develop a predictive visual intelligence source technology that can understand spatial situations and predict possible situations in advance based on multiple images and videos.

- SAIT Introduction Video** Oct. 2024
An interview about life as a researcher at SAIT. [\[link\]](#)
- SNU AI Summer School** Aug. 2022
20-minute presentation on “Graph-based Representation Learning for Class Discriminative Feature Embedding”. [\[link\]](#)

SCHOLARSHIPS AND HONORS	Samsung SAIT Scholarship	Jan. 2019 – Feb. 2022
	Awarded a scholarship from Samsung as part of an industry-academia collaboration program, receiving research funding during the Ph.D. program.	
	Best Presentation Award, Samsung AI Forum	Sep. 2018
	Recognized for outstanding presentation in the poster session and awarded a Galaxy Note 9 as a prize. [link]	
	Kwanjeong Educational Foundation Scholarship	Mar. 2013 – Feb. 2015
	Awarded a prestigious scholarship selecting top talent in South Korea, providing full tuition and financial support for two years.	
	SNU Eminence Scholarship	Sep. 2012 – Feb. 2013
	Awarded a full tuition scholarship in recognition of academic excellence.	
	SNU Superior Academic Performance Scholarship	Sep. 2011 – Feb. 2012
	Awarded a partial tuition scholarship in recognition of academic excellence.	
POSITION OF RESPONSIBILITIES	Lab Leader	Nov. 2019 – Oct. 2020
	In charge of project scheduling, job distributing, and coordinating feedback of research members.	
TEACHING EXPERIENCE	Topics in Control and Automation	Mar. 2020 – Jun. 2020
	Teaching Assistant for Prof. Jin Young Choi, Seoul National University. Delivered lectures on “Introduction to Graph Spectral Theory and Graph Convolutional Networks”.	
	Pattern Recognition	Mar. 2018 – Jun. 2018
	Teaching Assistant for Prof. Jin Young Choi, Seoul National University	
ACADEMIC ACTIVITIES	Reviewer	
	Serve as a reviewer for CVPR, ICCV, ECCV, NeurIPS, ICLR, SIGGRAPH, etc.	
	Research Assistant	Mar. 2016 – Aug. 2016
	Automation and Systems Research Institute , Seoul National University.	
	Visiting Student	Jan. 2016 – Feb. 2016
	Collaborative research at Robotics Lab , The University of Auckland , Auckland, New Zealand.	
	Exchange Student	Sep. 2014 – Feb. 2015
	Graz University of Technology (TU Graz), Graz, Austria.	

(Last update: February 20, 2025)